

MAT457 Lecture Notes

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1 Day 1: Here we go again! (Sep. 9, 2026)

Nabutovsky's office is in BA6104. Exams will be “difficult”, and you will be responsible for statements and proofs of theorems. You should know the meaning of the material. Unlike most Nabutovsky courses, tests will not be based on homework questions.

Undergraduate students are expected to know the basics of metric spaces, topological spaces, and set cardinality. You're also expected to have some exposure to the Lebesgue measure on $\mathbb{R}$ and $\mathbb{R}^n$, along with some of its properties.

Nabutovsky proceeds to survey the class about what they know. Here are the things he's asked (as to who is familiar on these).

1. Lebesgue density theorem in $\mathbb{R}^n$.
2. Fubini's theorem.
3. L^p spaces, and how to prove that they are complete.
4. Why care about the Lebesgue integral when the Riemann integral exists? (Per Nabutovsky, most functions you deal with in PDEs and other subjects are smooth, for which the Riemann integral will suffice. So why care?)

“Now I know where the class stands, and I will explain all these to you in due course.” In this course, we will extend the theory of Lebesgue integration to abstract spaces. Nabutovsky believes in starting slowly (with the later part of the course being much faster). We will start with the Lebesgue measure on $\mathbb{R}$.

Consider the space $X = \mathbb{R}$. Let us define a function $f: \mathcal{P}(\mathbb{R}) \rightarrow [0, +\infty]$ (observe that $+\infty$ is allowed). We want to make f a *measure*; in order to do that, it must satisfy

1. $f(\emptyset) = 0$,
2. (Countable additivity). If $E_1, E_2, \dots$ is a countable collection of disjoint sets, then

$$f\left(\bigcup_i E_i\right) = \sum_i f(E_i).$$

In particular, it follows from the above that if some E_i is of infinite measure, then $\bigcup_i E_i$ is also of infinite measure.

The following two properties apply only to the case when $X = \mathbb{R}$.

3. $f([0, 1]) = 1$,
4. f is translation-invariant, i.e., $f(E + t) = f(E)$, for all $E \in \mathcal{P}(\mathbb{R})$ and all $t \in \mathbb{R}$. Note that the set $E + t$ is given by $E + t = \{x \mid x - t \in E\}$.

At this point, Nabutovsky states that properties (iii) and (iv) are “kind of arbitrary”. Nabutovsky also said that there is in fact no function f that satisfies all 4 of these properties.

Proof. Consider the set $\mathbb{R}/\mathbb{Q}$. For convenience, let us restrict X down to $(0, 1)$; we say that $x \sim y$ if $y = x + q$, where $q \in \mathbb{Q}$, where the special addition symbol is defined by

$$a \circ b = \begin{cases} a + b & \text{if } a + b \in (0, 1), \\ a + b - 1 & \text{if } a + b \geq 1. \end{cases}$$

Let E consist of exactly one element from each of the equivalence classes of $X/\mathbb{Q}$ (from the axiom of choice)¹, for which the cardinality of E is $\mathbb{R}$. Now, define $E_q := E + q$ for $q \in \mathbb{Q} \cap [0, 1)$; then, for $q_1 \neq q_2$, we see that $E_{q_1} \cap E_{q_2} = \emptyset$, since, for all $t_1, t_2 \in E$ satisfying $t_1 + q_1 = t_2 + q_2$, it must be that

$$t_1 = t_2 + (q_2 - q_1) \implies t_1 \sim t_2,$$

so $t_1 = t_2$. Now,

$$0 < \mu((0, 1)) = \sum_{q \in \mathbb{Q} \cap [0, 1)} \mu(E_q).$$

By translation-invariance, it must be that $\mu(E_{q_1}) = \mu(E_{q_2})$, so we have two cases to consider: either $\mu(E_q) = 0$ (for any $q \in \mathbb{Q} \cap [0, 1)$), or it is greater than 0. In either case, we would get that the sum is 0 or $+\infty$, which contradicts property (iii). $\square$

At this point, Nabutovsky gave an example of another measure.

Definition 1 (Counting measure). *The counting measure counts the number of elements in a set; if there are infinitely many elements, the measure is infinite; if the set is finite, then the measure is exactly the number of elements in the set, per its definition.*

Definition 2 (Algebra). *Let X be a metric (or a topological space). Let $M \subset \mathcal{P}(X)$ (with $M \neq \emptyset$). We say M is an algebra if:*

1. *If $E \in M$, then its complement E^c is in M .*
2. *If $E_1, \dots, E_n \in M$, then their union $\bigcup_{i=1}^n E_i$ is in M .*

Definition 3 (σ -algebra). *In addition to the above properties, we ask that the union of any countable collection of elements of M is also in M .*

Using the fact that $A \cap B = (A^c \cup B^c)^c$, we also get that the countable intersection of elements of M is also in M .

¹note: this is a Vitali set

2 Day 2 - σ -algebras, measures (Sep 14, 2026)

Last time we defined the notion of a σ -algebra. Let us begin with some examples:

1. $\mathcal{M} = \{\emptyset, X\}$.
2. $\mathcal{M} = \mathcal{P}(X)$.
3. $\mathcal{M} = \{A \subseteq X : A \text{ or } A^c \text{ is countable}\}$.²

Suppose that we have some collection of subsets $A_\alpha \subseteq X$ and would like to find some σ -algebra $\mathcal{M}$ so that $A_\alpha \in \mathcal{M}$ for all α . First, observe that if $\mathcal{M}_\beta$ is some collection of σ -algebras on X and $A, A_1, A_2, \dots \in \mathcal{M}_\beta$ for all β , then for each β $A^c \in \mathcal{M}_\beta$ and $\bigcup_{j=1}^\infty A_j \in \mathcal{M}_\beta$, as $\mathcal{M}_\beta$ is a σ -algebra, and thus $\bigcap_\beta \mathcal{M}_\beta$ is also a σ -algebra. In particular, the intersection of all σ -algebras containing all A_α is also a σ -algebra containing all the A_α .³ Indeed this is also the smallest σ -algebra containing all of the A_α . We call this construction the σ -algebra generated by the A_α .

If X is a metric (or topological) space, we define the Borel σ -algebra of X to be the σ -algebra generated by the open subsets. A Borel set is an element of the Borel σ -algebra. If X is, for instance, separable,⁴ then the Borel σ -algebra is generated by the open balls.⁵ This is not the case in general. Consider $X = \mathbb{R}$ with the discrete metric $d(x, y) = 1$ if and only if $x \neq y$ and 0 otherwise. In this case, the σ -algebra generated by all open sets is clearly $\mathcal{P}(\mathbb{R})$, but the σ -algebra generated by the open balls is simply $\{\emptyset, \mathbb{R}\}$.

Let us devote some attention to the Borel σ -algebra of $\mathbb{R}$.

Proposition 4. *The Borel σ -algebra of $\mathbb{R}$ is generated by any of the following:*

1. *The set of all open intervals (a, b) .*
2. *The set of all closed intervals $[a, b]$.*
3. *The set of all half-open intervals $[a, b)$.*
4. *The set of all open rays (a, ∞) .*
5. *The set of all closed rays $(-\infty, a]$.*

Proof. $(2 \subseteq 1)$ $[a, b] = \bigcap_{n=1}^\infty (a - 1/n, b + 1/n)$. $(3 \subseteq 2)$ $[a, b) = \bigcup_{n=1}^\infty [a, b - 1/n]$. $(4 \subseteq 3)$ $[a, \infty) = \bigcup_{n=1}^\infty [a, n)$ and $(a, \infty) = \bigcup_{n=1}^\infty [a + 1/n, \infty)$. $(5 \subseteq 4)$ $(-\infty, a] = (a, \infty)^c$. $(1 \subseteq 5)$ $(a, b] = (-\infty, b] \cap (-\infty, a]^c$. $(a, b) = \bigcup_{n=1}^\infty (a, b - 1/n]$. And so the σ -algebras generated by any of (1)-(5) are equal. Since the Borel σ -algebra of $\mathbb{R}$ is, by the above remark, generated by the open balls, we are done. $\square$

Detour: what exactly are the Borel subsets of $\mathbb{R}$? Clearly the Borel σ -algebra contains all open, closed, G_δ , and F_σ sets, but it also contains all countable intersections of F_σ sets. Let us introduce

²It is clear that $\emptyset \in \mathcal{M}$ and that $\mathcal{M}$ is closed under complementation. Suppose $A_1, \dots \in \mathcal{M}$. If all the A_j are countable then $\bigcup_{j=1}^\infty A_j$ is countable and we are done. Otherwise, A_k^c is countable for some k , but then $(\bigcup_{j=1}^\infty A_j)^c = \bigcap_{j=1}^\infty A_j^c \subseteq A_k^c$, and so is countable.

³This collection is non-empty as, for instance, $\mathcal{P}(X)$ is a σ -algebra containing all the A_α .

⁴According to Nabu, we will almost always assume this to be the case.

⁵It is easy to check that, in this case, every open set is the countable union of open balls.

some notation. Let us say that A is a Π_n set if it can be written of the form

$$\bigcap_{i_1=1}^{\infty} \bigcup_{i_2=1}^{\infty} \cdots \left(\bigcup \bigcap \right)_{i_n=1}^{\infty} A_{i_1, i_2, \dots, i_n}$$

where the $A_{i_1, i_2, \dots, i_n}$ are either open or closed, i.e. starting⁶ with an intersection then alternating between intersection and union a total of n times. Similarly, let us say that A is a Σ_n set if it can be written of the form

$$\bigcup_{i_1=1}^{\infty} \bigcap_{i_2=1}^{\infty} \cdots \left(\bigcup \bigcap \right)_{i_n=1}^{\infty} A_{i_1, i_2, \dots, i_n}$$

where again the $A_{i_1, i_2, \dots, i_n}$ are either open or closed. Is every Borel subset of $\mathbb{R}$ a Π_n or Σ_n for some n ? No! Indeed, this recipe can be generalized to arbitrary countable ordinals⁷ He then alluded to the continuum hypothesis. My take on the point we was trying to make is how we may use transfinite induction to “explicitly” determine all of the Borel subsets by iterating the above process over the countable ordinals, a procedure which “terminates at the first uncountable ordinal.”⁸ Nabu also made the observation that the cardinality of the Borel σ -algebra of $\mathbb{R}$ is the cardinality of the continuum, so that’s something.

Later, we will often prefer to work with disjoint unions as opposed to arbitrary unions. This is where “disjointification” comes in. Given $A_n \in \mathcal{M}$, we may define $B_1 = A_1$ and $B_{n+1} = A_{n+1} \cap \bigcap_{j=1}^n A_j^c$. Note that the B_m are disjoint, lie in $\mathcal{M}$, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{m=1}^{\infty} B_m$.

Suppose we have some collection of measurable spaces⁹ $(X_\alpha, \mathcal{M}_\alpha)$ and want to define a σ -algebra on the product $\prod_\alpha X_\alpha$, which we’ll denote by¹⁰ $\bigotimes_\alpha \mathcal{M}_\alpha$. To do this, we mimic the construction of the product topology in topology, and take the σ -algebra generated by the preimages $\pi_\beta^{-1}(A)$ for $A \in \mathcal{M}_\beta$, where $\pi_\beta : \prod_\alpha X_\alpha \rightarrow X_\beta$ is the projection map. For example, consider $\mathbb{R} \times \mathbb{R}$ where we take each copy of $\mathbb{R}$ to have the Borel σ -algebra. Since the open intervals generate the Borel σ -algebra of $\mathbb{R}$, it follows¹¹ that the product σ -algebra is generated by the open rectangles $(a, b) \times (c, d)$. These in turn are the open balls of $\mathbb{R}^2$ under the metric $d_{\max}(x, y) = \|x - y\|_\infty = \max(|x_1 - y_1|, |x_2 - y_2|)$; since all norms on $\mathbb{R}^2$ are equivalent, the product σ -algebra is also generated by the usual open balls of $\mathbb{R}^2$, i.e. the product σ -algebra of $\mathbb{R} \times \mathbb{R}$ and the Borel σ -algebra of $\mathbb{R}^2$ coincide.

Definition 5 (Measure). *Let $(X, \mathcal{M})$ be a measurable space. A measure is a map $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that:*

1. $\mu(\emptyset) = 0$.
2. “Countable additivity” If $A_n \in \mathcal{M}$ are disjoint then $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

Let’s see some examples:

1. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = \infty$ if A has infinitely many elements and $\mu(A) = |A|$ (the cardinality of A) otherwise. More generally, given $f : X \rightarrow [0, \infty]$ we may define $\mu_f(A) = \sum_{a \in A} f(a)$.
2. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = 0$ if A is countable and ∞ otherwise.

⁶Reading left-to-right.

⁷An ordinal is an equivalence class of ordered sets.

⁸Someone once told me that transfinite induction was first used/formulated to “describe” the Borel subsets of $\mathbb{R}$; so be sure to memorize that for pub trivia.

⁹A measurable space is a non-empty set X equipped with a σ -algebra $\mathcal{M}$.

¹⁰:(.

¹¹Exercise!

3. Given a measure $\mu_0 : \mathcal{M}_0 \rightarrow [0, \infty]$ and $A \subseteq X$, define $\mathcal{M} = \{B \subseteq X : A \cap B \in \mathcal{M}_0\}$ and $\mu : \mathcal{M} \rightarrow [0, \infty]$ by $\mu(B) = \mu_0(A \cap B)$. “The restriction measure”
4. Given $x_0 \in X$, define $\mu(A) = 1$ if $x_0 \in A$ and 0 otherwise. “Dirac measure”
5. Lebesgue measure on $\mathbb{R}^n$.

The following is a non-example. Define $\mu(A) = \infty$ if A is infinite and 0 otherwise.

Definition 6 (Finite). *A measure μ is finite if $\mu(X) < \infty$.*

For example, $\mu(A) = \int_A e^{-x^2} dx$ on the σ -algebra of Lebesgue measurable subsets of $\mathbb{R}$, the Dirac measure. A non-example: the Lebesgue measure.

Definition 7 (σ -finite). *A measure μ is σ -finite if there are countably many measurable sets E_i so that $\mu(E_i) < \infty$ and $X = \bigcup_{i=1}^{\infty} E_i$.*

For example, the Lebesgue measure. A non-example: the counting measure on $\mathbb{R}$.

Definition 8 (Semi-finite). *A measure μ is semi-finite if $E \in \mathcal{M}$ and $\mu(E) = \infty$ implies there is some $F \subseteq E$ with $F \in \mathcal{M}$ and $0 < \mu(F) < \infty$.*

Exercise 9. *Show that σ -finite implies semi-finite, but that the converse may fail.*

Definition 10 (Measure space). *A measure space is a triple $(X, \mathcal{M}, \mu)$ where X is a non-empty set, $\mathcal{M}$ is a σ -algebra on X , and $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a measure.*

Theorem 11. *Let $(X, \mathcal{M}, \mu)$ be a measure space. We have:*

1. “Monotonicity” If $A, B \in \mathcal{M}$ and $A \subseteq B$ then $\mu(A) \leq \mu(B)$.
2. “Countable subadditivity” If $A_n \in \mathcal{M}$ then $\mu(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$.
3. “Continuity from below” If $E_1 \subseteq E_2 \subseteq \dots$ then $\mu(\bigcup_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_n)$.
4. “Continuity from above” If $E_1 \supseteq E_2 \supseteq \dots$, $\mu(E_1) < \infty$ then $\mu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_n)$.

Proof. For (1), note that B may be written as the disjoint union of A and $B \cap A^c$, so by additivity $\mu(B) = \mu(A) + \mu(B \cap A^c) \geq \mu(A)$, since μ is non-negative.¹² For (2), apply disjointification, countable additivity, and monotonicity to obtain $\mu(\bigcup_{n=1}^{\infty} A_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^{\infty} \mu(B_m) = \sum_{m=1}^{\infty} \mu(A_m \cap \bigcap_{j=1}^{m-1} A_j) \leq \sum_{m=1}^{\infty} \mu(A_m)$. For (3), take $B_1 = E_1$ and $B_{m+1} = E_{m+1} \setminus E_m$. Since the B_m are disjoint and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{m=1}^{\infty} B_m$, we have $\mu(\bigcup_{n=1}^{\infty} E_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^N \mu(B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(\bigcup_{m=1}^N B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(E_N) + \sum_{m=N+1}^{\infty} \mu(B_m)$, where the latter summand is the tail of a convergent series, and so vanishes in the limit. For (4), put $F_j = E_1 \setminus E_j$. Notice that $F_1 \subseteq F_2 \subseteq \dots$ and so by (3) we have $\mu(\bigcup_{j=1}^{\infty} F_j) = \lim_{j \rightarrow \infty} \mu(F_j)$. Now, $\mu(F_j) = \mu(E_1) - \mu(E_j)$ by additivity; and $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_{n=1}^{\infty} E_n$. All together, we have $\mu(E_1) - \mu(\bigcap_{n=1}^{\infty} E_n) = \mu(\bigcup_{j=1}^{\infty} F_j) = \mu(E_1) - \lim_{n \rightarrow \infty} \mu(E_n)$. Since $\mu(E_1) < \infty$, we conclude. $\square$

Note that the hypothesis in (4) that $\mu(E_1) < \infty$ is important. Consider the Lebesgue measure on $\mathbb{R}$ and $E_i = (i, \infty)$. Then $\bigcap_{i=1}^{\infty} E_i = \emptyset$ but $\lim_{i \rightarrow \infty} \mu(E_i) = \infty$.

A measurable set of measure zero is called a null set. By monotonicity, measurable subsets of null sets are null sets. However, it could be the case that a subset of a null set fails to be measurable. For a trivial example, take $X = \{a, b\}$, $\mathcal{M} = \{\emptyset, X\}$, and μ the counting measure.

¹²Later in the course, we will discuss signed measures, which are precisely what it says on the tin.

Definition 12 (Complete measure). *A measure is said to be complete¹³ if all subsets of null sets are measurable.*

Definition 13 (Completion). *Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\overline{\mathcal{M}}$ be the σ -algebra generated by $\mathcal{M}$ and the subsets of null sets. The (unique) extension of μ to $\overline{\mathcal{M}}$ is called the completion.*

Nabu claimed that $\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subseteq Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}$. He didn't have time for a proof, but I've provided one below. While technically it should be a theorem that the extension of a measure actually exists, this result makes it clear that setting $\overline{\mu}(A \cup N) = \mu(A)$ is well-defined. Uniqueness, along with some other checks, are straight-forward.¹⁴

Proposition 14. $\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subseteq Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}$.

Proof. $\emptyset$ is a null set, so $\emptyset = \emptyset \cup \emptyset \in \overline{\mathcal{M}}$. By countable subadditivity, the countable union of null sets is a null set, taken with the identity $\bigcup_{n=1}^{\infty} A_n \cup N_n = (\bigcup_{n=1}^{\infty} A_n) \cup (\bigcup_{n=1}^{\infty} N_n)$, it quickly follows that $\overline{\mathcal{M}}$ is closed under countable unions. Now suppose $E = A \cup N \in \overline{\mathcal{M}}$ and pick $Z \in \mathcal{M}$ with $N \subseteq Z$ and $\mu(Z) = 0$. Now, $N \subseteq Z$ implies $Z^c \subseteq N^c$ and so $N^c = Z^c \cup (Z \setminus N)$. Substituting this in, we have $E^c = A^c \cap N^c = (A^c \cap Z^c) \cup (A^c \cap (Z \setminus N))$. But $A^c, Z^c \in \mathcal{M}$ and $A^c \cap (Z \setminus N) \subseteq Z$ which has measure zero, and thus E^c is of the desired form. $\square$

¹³This terminology has nothing to do with the notion of completeness for metric spaces.

¹⁴Exercise!